

Updating parameters

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Composing layers

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Detect overshooting for Updating parameters. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Update a new model independently

Use $x = 1$, target 4, $w = 1$, $b = 1$, squared loss and learning rate 0.1. Compute prediction, residual, gradient [2 times residual times x , 2 times residual], new parameters and new loss.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Apply a negative gradient

Starting with parameter $a = 2$, gradient -3 and learning rate 0.2, calculate the new a .

My reasoning and evidence

6. W2 Detect overshooting

For the lesson start [1, 0], apply rate 0.3 and calculate prediction and loss.

My reasoning and evidence

7. W3 Preserve one snapshot

Explain why recomputing the bias gradient after changing w is not the simultaneous update specified here.

My reasoning and evidence

8. W4 Update a new model independently

Use $x = 1$, target 4, $w = 1$, $b = 1$, squared loss and learning rate 0.1. Compute prediction, residual, gradient [2 times residual times x , 2 times residual], new parameters and new loss.

My reasoning and evidence

9. W5 Evaluate a zero gradient

At a point with gradient $[0, 0]$, what does this update do? Does that alone prove a global minimum on any loss surface?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled updating parameters evidence sheet to Image-pattern Network Investigation. Show how you can apply a learning rate to an ordered gradient. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Updating parameters

Use one explicit gradient update and then recompute the loss.

Model $y=2w+b$, target 5. Start $w=1$, $b=1$; learning rate 0.05. Initial $y=3$, error $y-\text{target}=-2$, squared loss 4.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update

squared error

before



after



Bar length shows the supplied value; the marked baseline is zero.

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for $[w, b] = [2 \times (-2) \times 2, 2 \times (-2)] = [-8, -4]$.



2. Update simultaneously

New $w = 1 - 0.05 \times (-8) = 1.4$; new $b = 1.2$.



3. Check new loss

New $y = 4$, residual -1 , loss 1 : lower than 4 .

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use one explicit gradient update and then recompute the loss.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? New $y=4$, residual -1 , loss 1 : lower than 4 .

1. Check new loss
2. Compute gradient
3. Update simultaneously

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check new loss New $y=4$, residual -1 , loss 1 : lower than 4 . Gradient descent subtracts learning rate times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original $[w, b]$ before updating either one. The small step brings the prediction closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new prediction and loss must be recalculated.

Practice 2: explain the deciding evidence

Changed-case practice: Set learning rate 0.2 from the same starting point. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. New $w=2.6$, $b=1.8$, $y=7$, squared loss 4 : no improvement.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: New $w=2.6$, $b=1.8$, $y=7$, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Repair a missing building block

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer inputs. Short practice: What intermediate value explains the changed output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
